

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1001

Orchid 69 kV
June 03, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1001. The Interconnection Customer proposes a 1091.5 MW natural gas combined cycle facility interconnecting at the Orchid 69 kV substation in the North Carolina area. The request was received on May 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 422 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$109,848,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tecumseh Energy Center - Honeyville 138 KV line	Mecklenburg Power Station - Kerr Dam 345 KV L	182.5	297	19.8
Tecumseh Energy Ctr 345/138 kV Autotransformer	Kerr Dam - Mecklenburg Power Sta 345 KV line	157.9	349	19.8
Quindaro - Highlands Windfarm 138 KV L	Ols Energy Chino - Pine River Alte 138 KV line	136.3	223	33.9
Huben - Highlands Windfarm 138 KV	Pine Riv Alte - Ols Energy Chino 138 KV	135.2	220	33.9
Quindaro - Savanna 138 KV	Pine River Alte - Ols Energy Chino 138 KV	135.1	229	33.9
Cape Horn - Huben 138 KV L	Kerr Dam - Mecklenburg Power Sta 345 KV	132.4	234	5.2
Kerr Dam 345/138kV TX	Gaston - Tecumseh Energy Ctr 345 KV	131.3	507	5.5
Grady 138/69kV Transformer T1	Grady 138 kV/69 kV Autotransformer Unit 3	123.1	107	17.1
Grady 138-69 kV Auto	Grady Substation 138-69 kV Autotransformer	123.1	107	17.1
Grady 138 kV/69 kV Transformer Bank Bank 3	Grady 138/69kV Transformer T1	123.1	107	17.1
Pine Riv Alte - Ols Energy Chino 138 KV line	Quindaro - Savanna 138 KV	112.5	376	48.9
E Sparta 138/69kV Transformer	Tecumseh Energy Ctr - Gaston 345 KV	100.6	99	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tecumseh Energy Center - Honeyville 138 KV line	Rebuild line with higher-capacity conductor	\$10,034,000	85% (\$8,543,000)
Tecumseh Energy Ctr 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$16,185,000	85% (\$13,778,000)
Quindaro - Highlands Windfarm 138 KV L	Rebuild line with higher-capacity conductor	\$6,554,000	100% (\$6,554,000)
Huben - Highlands Windfarm 138 KV	Rebuild line with higher-capacity conductor	\$9,572,000	100% (\$9,572,000)
Quindaro - Savanna 138 KV line	Rebuild line with higher-capacity conductor	\$11,063,000	100% (\$11,063,000)
Cape Horn - Huben 138 KV L	Rebuild line with higher-capacity conductor	\$16,020,000	86% (\$13,695,000)
Kerr Dam 345/138kV TX	Replace transformer with higher-rated unit	\$17,878,000	60% (\$10,781,000)
Grady 138/69 kV Autotransformer #1	Replace transformer with higher-rated unit	\$5,326,000	100% (\$5,326,000)
Grady 138-69 kV Auto	Replace transformer with higher-rated unit	\$4,141,000	100% (\$4,141,000)
Grady 138/69kV TX	Replace transformer with higher-rated unit	\$6,096,000	100% (\$6,096,000)
Ols Energy Chino - Pine Riv Alte 138 KV	Rebuild line with higher-capacity conductor	\$12,980,000	100% (\$12,980,000)
E Sparta 138 kV/69 kV Autotransformer	Replace transformer with higher-rated unit	\$7,319,000	100% (\$7,319,000)
	Total allocated to project		\$109,848,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.